

Requirements Elicitation: Finding Out What Your Users Need

Discussion About the Reading

- Why do we need nonfunctional requirements?
- What is requirements validation and how might we do it?
- Compare & contrast user requirements, system requirements, and design specifications.
- What should you do when requirements conflict?

A Difficulty

- You have 10 weeks for the project
- Many "real" projects are much bigger (more resources)
- There are lots of aspects of "real" projects that you can't do in class
- Lecture will include some techniques you don't have time to use now

Eliciting Requirements

- We want to meet a **real** user's needs
- What arguments should win?
 - Not the loudest ones
 - Instead, ones that are grounded in the best data.
- HCI mantra: "The user is not like me"

Interviews

- Consider all stakeholders
- Who are the stakeholders of your system?

Interview Structure

- *Semistructured*: Ask questions in a conversational way (potentially out of order)
- Your results will depend on interviewer skill.

Focus Groups

- Focus group: gather 5-7 or 8-12 participants for a group interview
- Pro: more participants, less experimenter time
- Discussions reveal similarities and differences
- Con: quiet people might not get heard
- Skill is needed to manage conversation
- Analysis can be tricky (interruptions, changes of speaker)

Demonstration: Writing Questions

- "A taco is a kind of sandwich, right?"
- Is a taco a sandwich?
- What food categories are the items in the picture in?
- What do you expect to see on the menu at a sandwich restaurant?
- What makes something a sandwich?



Demonstration (2)

- How do you feel about covariant return types?
 - Use terms your participant knows.
- When do you usually decide to start using the debugger?
 - Think of the last bug you fixed. What debugging strategies did you use?

Designing Questions

- Neutral: unbiased, nonjudgmental
- Simple
- Open-ended
- Speak their language
- Ask for demonstrations or recall of concrete events

Simple Questions

- "What were the strengths and weaknesses of the compiler and IDE?"
 - -> "What did you think of the compiler" & "What did you think of the IDE?"

Netural Questions

- "You find homework pretty stressful, right?"
 - -> "What kinds activities do you find the most stressful?"
- "Why do you like this design?"
 - What if they didn't like the design?

Recording Data

- Write notes
 - Rewrite and summarize after the interview
- Record audio & transcribe
- Screen capture (if there are demonstrations)

Rapport

- Be nonjudgmental — develop a poker face!
- Keep people comfortable. Water? Snacks?

Conducting the Interview

- Start with easy questions
- Listen!
- Provide opportunities to continue: "Is there anything else you wanted to tell me?"
- Ask for clarification when needed: "What exactly do you mean when you say...?"

Activity

- You want to know how people remember things that need to be done (you hope to create a new calendar/to-do system).
- Write two *open-ended* questions asking about how people organize their tasks.
- Ask them of your neighbor. Summarize their answers.

<https://www.gradescope.com/courses/940938/assignments/>

Contextual Inquiry

- Observe people doing the work as it is done now
- For your project: you can find worksheets, lesson plans, etc. online
- Printing example

Goal-Based Approaches

- Analyze the problem domain
- Example: *automate all aspects of baggage handling from passenger origin to destination.*
 - Goal 1: Automate the tracking of baggage from check-in to pick-up.
 - Goal 2: Automate the routing of baggage from check-in counter to plane.
 - Goal 3: Reduce the amount of lost luggage to 1%.

Prototyping

- Prototype user interface and interview users
- We'll talk about prototyping later

Scenarios

- Informal descriptions of the system in use

A customer walks into the pet store and fills the cart with a variety of items...the cashier scans products using a bar code reader. As each item is scanned, the sale is totaled and the inventory is appropriately updated. Upon completion of product scanning a subtotal is computed. Then any coupons and discounts are entered. A new subtotal is computed and applicable taxes are added. A receipt is printed and the customer pays using cash, credit card, debit card, or check. All appropriate totals (sales, tax, discounts, rebates, etc.) are computed and recorded.

In-Class Activity (ICA)

- WebReg is old, antiquated, and needs to be replaced.
- You're in charge.
- What will you do to elicit requirements?
- Give two things you will do. Explain (briefly) why each is appropriate.